HOMEWORK 7

ANTONY PEREZ

3.1. If A is an $n \times n$ matrix, how are the determinants $\det A$ and $\det(5A)$ related?
Remark: $\det(5A) = 5 \det A$ only in the trivial case of 1×1 matrices.

Solution. Let $\det(A) = \alpha$. These two are related because $\det(5A) = 5^n \times \alpha$. This is because we can pull out the scalar 5 out of each column vector, and, since there are n column vectors, we do this n times, giving us $5^n \det(A) = 5^n \times \alpha$. ■

3.2. How are the determinants $\det A$ and $\det B$ related if

(a)
$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \qquad B = \begin{pmatrix} 2a_1 & 3a_2 & 5a_3 \\ 2b_1 & 3b_2 & 5b_3 \\ 2c_1 & 3c_2 & 5c_3 \end{pmatrix};$$

(b)
$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \qquad B = \begin{pmatrix} 3a_1 & 4a_2 + 5a_1 & 5a_3 \\ 3b_1 & 4b_2 + 5b_1 & 5b_3 \\ 3c_1 & 4c_2 + 5c_1 & 5c_3 \end{pmatrix}.$$

Solution. For (a), they are related by linearity. For (b), they are also related by linearity and row/column addition. Pulling the scalar out of each column gives $\det B = 2 \cdot 3 \cdot 5 \det A = 30 \det A$ in (a). In (b), splitting column 2 gives $3 \cdot 4 \cdot 5 \det A$ plus a determinant with two proportional columns, which is 0, so $\det B = 60 \det A$. ■

3.3. Using column or row operations compute the determinants

$$\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{vmatrix}, \quad \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}, \quad \begin{vmatrix} 1 & 0 & -2 & 3 \\ -3 & 1 & 1 & 2 \\ 0 & 4 & -1 & 1 \\ 2 & 3 & 0 & 1 \end{vmatrix}, \quad \begin{vmatrix} 1 & x \\ 1 & y \end{vmatrix}.$$

Solution. Let A, B, C, D be the matrices above respectively. For A, swap R_1 with R_2, which flips the sign. Then $R_3 + 2R_1$. We finally have $r_3 - 3R_2$ We now multiply across the diagonal and get $-((-1)(1)(-12)) = -12$. $\det(A) = -12$.
For B, $R_2 - 4R_1; R_3 - 7R_1; R_3 - 2R_2$. This makes Row 3 all zeros, showing linear dependence, which shows $\det(B) = 0$.
For C, $R_2 + 3R_1; R_4 - 2R_1; R_3 - 4R_2; R_4 - 3R_2; R_4 - R_3$. This gives us a triangular matrix, so $\det(C) = 1 \cdot 1 \cdot 19 \cdot 5 = 95$.
For D, $R_2 - R_1$. $\det(D) = y - x$. ■

3.4. A square $(n \times n)$ matrix is called *skew-symmetric* (or *antisymmetric*) if $A^T = -A$. Prove that if A is skew-symmetric and n is odd, then $\det A = 0$. Is this true for even n?

Proof. Since $A^T = -A$, taking determinants gives

$$\det(A) = \det(A^T) = \det(-A) = (-1)^n \det(A).$$

Since n is odd, $\det(A) = -\det(A)$, so $\det(A) = 0$. This is not true for even n. $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is skew-symmetric, but $\det(A) = 1$. ■

3.5. A square matrix is called *nilpotent* if $A^k = \mathbf{0}$ for some positive integer k. Show that for a nilpotent matrix A, $\det A = 0$.

Proof. Since A is nilpotent, there exists $k \in \mathbb{N}$ such that $A^k = \mathbf{0}$. By Theorem 3.5, we get

$$(\det A)^k = \det(A^k) = \det(\mathbf{0}) = 0.$$

This implies $\det A = 0$. ■

3.6. Prove that if the matrices A and B are similar, then $\det A = \det B$.

Proof. Let $A = PBP^{-1}$. $\det(A) = \det(PBP^{-1}) \to \det(PP^{-1})\det(B) \implies \det A = \det B$. ■

3.7. A real square matrix Q is called *orthogonal* if $Q^TQ = I$. Prove that if Q is an orthogonal matrix then $\det Q = \pm 1$.

Proof. Let $Q^TQ = I$. $\det(Q^TQ) = \det(I) \implies \det(Q^T)\det(Q) = 1$. Since $\det(Q^T) = \det(Q)$, they must have the same value. Since they're being multiplied together, the only solutions to this equation is if $\det(Q^T) = \det(Q) = \pm 1$. This concludes the proof. ■

3.8. Show that

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (z-x)(z-y)(y-x).$$

This is a particular case of the so-called Vandermonde determinant.

Proof. Subtracting the first row from the second and third rows gives

$$\det(A) = \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix}.$$

Factoring out $y - x$ and $z - x$, then subtracting the second row from the third, gives

$$\det(A) = (y-x)(z-x)\begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 0 & z-y \end{vmatrix}.$$

Multiplying across the diagonal, we get

$$\det(A) = (z-x)(z-y)(y-x).$$

■

3.9. Let points A, B and C in the plane $\mathbb{R}^2$ have coordinates (x_1, y_1), (x_2, y_2) and (x_3, y_3) respectively. Show that the area of triangle ABC is the absolute value of

$$\frac{1}{2}\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}.$$

Hint: use row operations and geometric interpretation of 2×2 determinants (area).

Proof. Subtracting the first row from the second and third rows gives

$$\det(A) = \begin{vmatrix} 1 & x_1 & y_1 \\ 0 & x_2 - x_1 & y_2 - y_1 \\ 0 & x_3 - x_1 & y_3 - y_1 \end{vmatrix} = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix}.$$

The absolute value of this determinant gives the area of the parallelogram determined by the vectors from A to B and C. Since the triangle has half this area,

$$\text{Area}(ABC) = \frac{1}{2}\left|\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}\right|.$$

■

3.10. Let A be a square matrix. Show that block triangular matrices

$$\begin{pmatrix} I & * \\ \mathbf{0} & A \end{pmatrix}, \qquad \begin{pmatrix} A & * \\ \mathbf{0} & I \end{pmatrix}, \qquad \begin{pmatrix} I & \mathbf{0} \\ * & A \end{pmatrix}, \qquad \begin{pmatrix} A & \mathbf{0} \\ * & I \end{pmatrix}$$

all have determinant equal to $\det A$. Here $*$ can be anything.

Solution. All four matrices are block triangular, so their determinants equal the products of the determinants of their diagonal blocks. Since $\det(I) = 1$, each determinant is $\det(A)$. ■

The following problems illustrate the power of block matrix notation.

3.11. Use the previous problem to show that if A and C are square matrices, then

$$\det \begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \det A \det C.$$

Hint: $\begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \begin{pmatrix} I & B \\ \mathbf{0} & C \end{pmatrix} \begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix}$.

Proof. Using the factorization in the hint and taking determinants, we get

$$\det \begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \det \begin{pmatrix} I & B \\ \mathbf{0} & C \end{pmatrix} \det \begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix}.$$

By Problem 3.10, these determinants are $\det(C)$ and $\det(A)$, respectively. Therefore,

$$\det \begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \det(A) \det(C).$$

■

3.12. Let A be $m \times n$ and B be $n \times m$ matrices. Prove that

$$\det \begin{pmatrix} 0 & A \\ -B & I \end{pmatrix} = \det(AB).$$

Hint: While it is possible to transform the matrix by row operations to a form where the determinant is easy to compute, the easiest way is to right multiply the matrix by $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}$.

Proof. Right multiplying by the matrix in the hint gives

$$\begin{pmatrix} 0 & A \\ -B & I \end{pmatrix} \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \begin{pmatrix} AB & A \\ 0 & I \end{pmatrix}.$$

Taking determinants and applying Problems 3.10 and Theorem 3.5, we get

$$\det \begin{pmatrix} 0 & A \\ -B & I \end{pmatrix} = \det(AB),$$

since $\det \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = 1$ and $\det \begin{pmatrix} AB & A \\ 0 & I \end{pmatrix} = \det(AB)$. ■